```
// IoT Based Plant Monitoring System
// NodeMCU ESP8266 + DHT11 + Soil Moisture + Relay + Blynk
#define BLYNK_TEMPLATE_ID "TMPLxxxxxxx"
#define BLYNK_TEMPLATE_NAME "Plant Monitor"
#define BLYNK_AUTH_TOKEN "Your_Auth_Token"
#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
#include <DHT.h>char ssid[] = "Your_WiFi_Name";char pass[] = "Your_WiFi_Password";
#define DHTPIN D4
#define DHTTYPE DHT11
#define SOIL_PIN A0
#define RELAY_PIN D5// Calibrate these two values for the actual soil sensor.int
DRY_VALUE = 850;  int WET_VALUE = 350;
// Many relay modules are active LOW.// Change to HIGH if your relay is active HIGH.
#define RELAY_ON LOW
#define RELAY_OFF HIGHDHT dht(DHTPIN, DHTTYPE);
BlynkTimer timer;  bool manualPump = false;  bool autoMode = true;
void sendSensorData(){ float temperature = dht.readTemperature();
 float humidity = dht.readHumidity();
int rawSoil = analogRead(SOIL_PIN);
if (isnan(temperature) || isnan(humidity))
{ Serial.println("DHT11 read failed."); return; }
 int soilPercent = map(rawSoil, DRY_VALUE, WET_VALUE, 0, 100);
 soilPercent = constrain(soilPercent, 0, 100);
 Blynk.virtualWrite(V0, temperature);
Blynk.virtualWrite(V1, humidity);
Blynk.virtualWrite(V2, soilPercent);
// Automatic irrigation
 if (autoMode) { if (soilPercent < 35) { digitalWrite (RELAY_pin, relay_on }
else if (soilPercent > 55) { digitalWrite(RELAY_PIN, RELAY_OFF); }
```

```
Serial.print(" % | Soil: ");
Serial.print(soilPercent);
 Serial.println(" %");}
// V3 = manual pump switch in BlynkBLYNK_WRITE(V3)
{ manualPump = param.asInt();
 if (!autoMode) { digitalWrite(RELAY_PIN, manualPump ? RELAY_ON : RELAY_OFF); }}
// V4 = automatic/manual mode switchBLYNK_WRITE(V4)
{ autoMode = param.asInt(); if (autoMode)
{ digitalWrite(RELAY_PIN, RELAY_OFF); }
 else { digitalWrite(RELAY_PIN, manualPump ?
 RELAY_ON : RELAY_OFF);
Serial.print("Temperature: ");
Serial.print(temperature); Serial.print(" C | Humidity: "); Serial.print(humidity);
void setup()
{ Serial.begin(115200);
pinMode(RELAY_PIN, OUTPUT);
digitalWrite(RELAY_PIN, RELAY_OFF);
dht.begin();
 Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
timer.setInterval(2000L, sendSensorData);}
void loop(){ Blynk.run(); timer.run();
```